

Malware Analysis Lab Assessment-5

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Q1) Provide a single screenshot showing your VM snapshot manager or VM settings and a short

line confirming network isolation and snapshot taken. State x64dbg version used.

```
Administrator: Windows PowerShell
PS C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91 > certutil -hashfile C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91\random.exe SHA256
SHA256 hash of C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91\random.exe:
87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91
CertUtil: -hashFile command completed successfully.
PS C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91 > certutil -hashfile C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91\random.exe MD5
MD5 hash of C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91\random.exe:
f5072721dc30dac90de5ebfee5f55316
CertUtil: -hashFile command completed successfully.
PS C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91 > dir C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91\random.exe

Directory: C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91

Mode                LastWriteTime         Length Name
----                -
a-----          11/7/2025   6:37 AM           3912992 random.exe

PS C:\Users\Tharun\Downloads\87efcded6a04790242c61a043b55375c7cfbdf2e500806fc247cd29fb8756e91 >
```

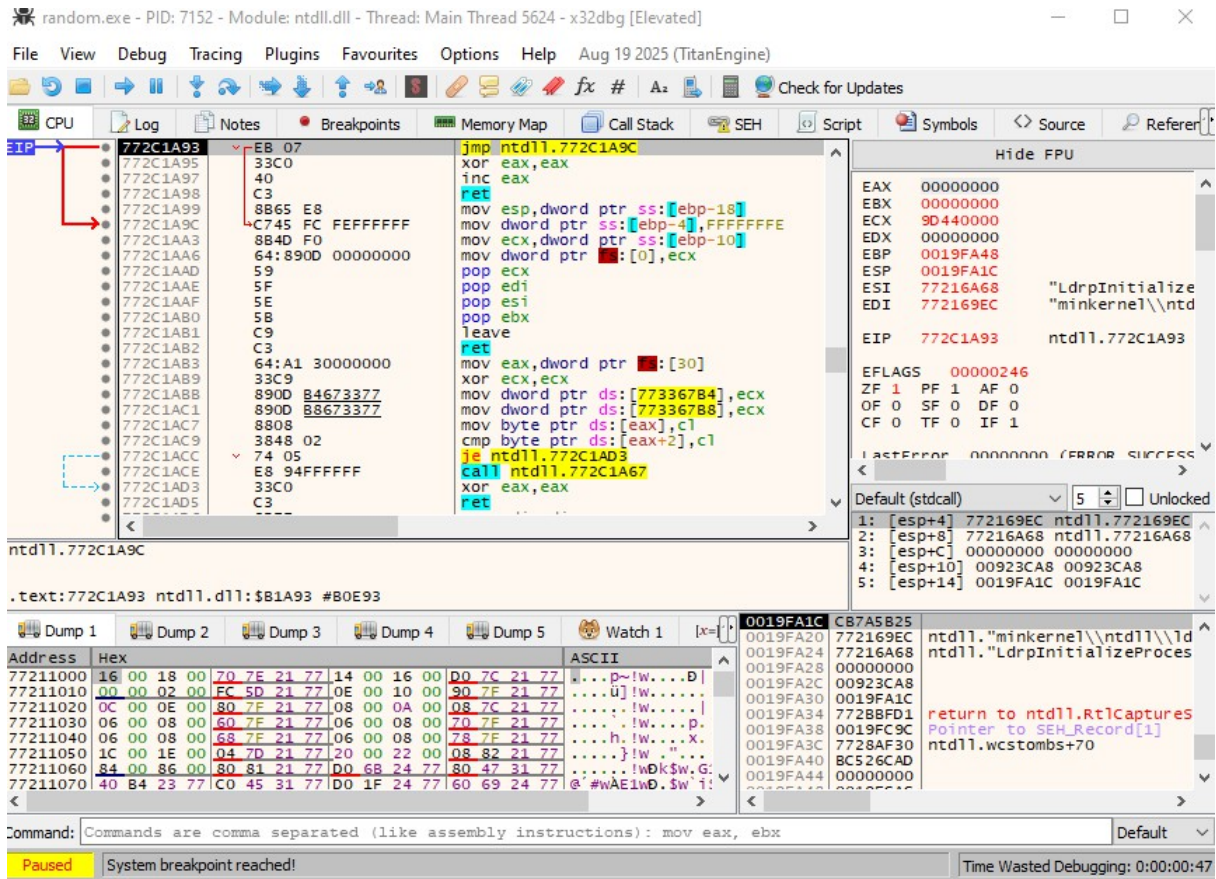
Q2) Produce a one-paragraph static summary: file size, MD5/SHA256 hash, PE characteristics

(32/64-bit, imports visible), and any obvious strings you observed briefly with a strings tool.

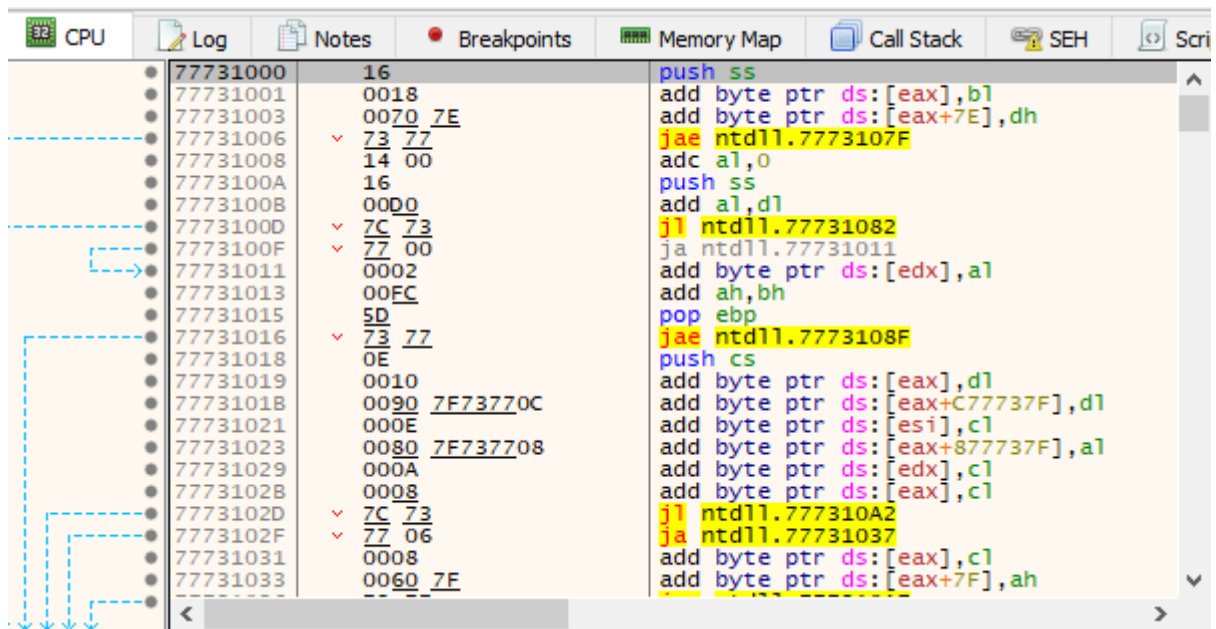
No deep static reversing required — just a concise baseline.

Ans) Start x64dbg and load the sample, Launch x64dbg File → Open → select random.exe. x64dbg will stop at the module load/entry point before execution.

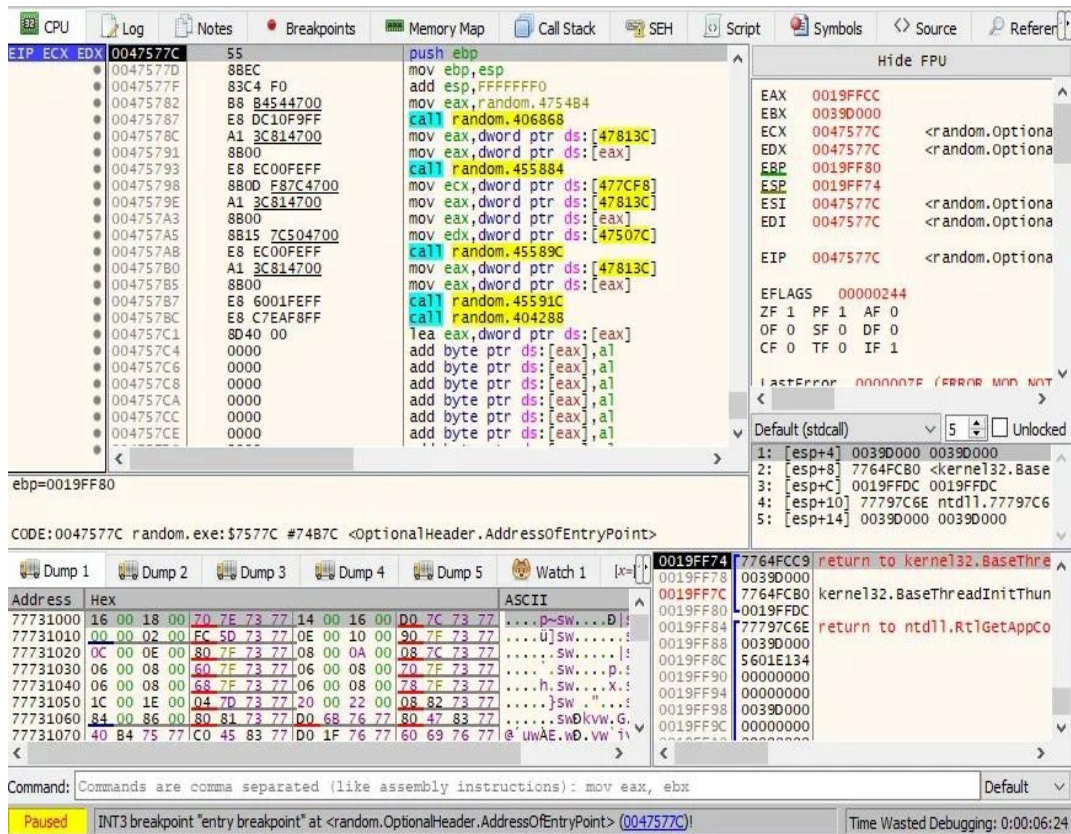
Screenshots are attached below



Entry Point before running the malware is 77731000



After running the malware: 0047577C



3) Analysis:

The entry point begins with a standard function prologue (push ebp; mov ebp, esp) to set up the stack frame, followed by a push eax to preserve the register.

The code immediately calls several API functions via indirect calls through the Import Address Table (IAT), with call random.404288 at offset +0x10 being the first significant call.

Multiple mov eax, dword ptr ds:[address] instructions suggest the malware is resolving or preparing function pointers from the IAT.

There are no obvious anti-debug checks visible in the first basic block; execution flows linearly through API setup calls before reaching conditional logic at lower addresses.

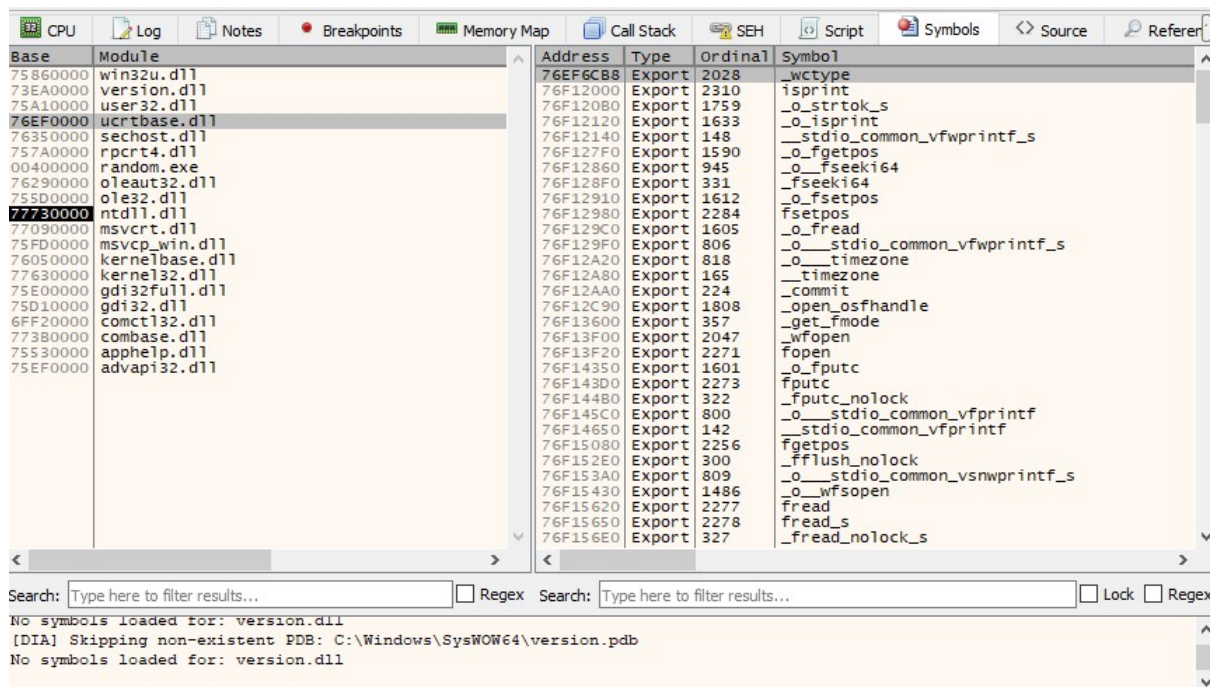
Q4) Run the sample under x64dbg (within VM). Set sensible breakpoints to observe runtime

behaviour (no command list required — describe what you set). Provide:

Ans) Prepare defensive breakpoints (anti-debug / flow observation)

Set breakpoints to catch early runtime behaviour. Use these as watchpoints to see what the sample does when run:

- i. Break on API resolution / dynamic import:
- ii. LoadLibraryA / LoadLibraryW — set breakpoint (right-click import view or use bp command).
- iii. GetProcAddress set a breakpoint. Memory allocation / protection (common unpacker behaviours): VirtualAlloc, VirtualProtect — set breakpoints.
- iv. Suspicious process manipulation / injection: WriteProcessMemory, CreateRemoteThread, OpenProcess set breakpoints. Thread creation / exit points and CreateThread, ExitProcess set breakpoints.



Setting breakpoint at GetProcAddress

CPU Log Notes Breakpoints Memory Map Call Stack SEH Script Symbols Source Referer

Base	Module	Address	Type	Ordinal	Symbol
75860000	win32u.dll				
73EA0000	version.dll	73EA5094	Import		kernelbase.GetProcAddress
75A10000	user32.dll				
76EF0000	ucrtbase.dll				
76350000	sechost.dll				
757A0000	rpcrt4.dll				
00400000	random.exe				
76290000	oleaut32.dll				
755D0000	ole32.dll				
77730000	ntdll.dll				
77090000	msvcrt.dll				
75FD0000	msvc_p_win.dll				
76050000	kernelbase.dll				
77630000	kernel32.dll				
75E00000	gdi32full.dll				
75D10000	gdi32.dll				
6FF20000	comctl32.dll				
77380000	combase.dll				
75530000	apphelp.dll				
75EF0000	advapi32.dll				

Search: Type here to filter results... ☐ RegEx Search: GetP ☐ Lock ☐ RegEx

No symbols loaded for: version.dll
[DIA] Skipping non-existent PDB: C:\Windows\SysWOW64\version.pdb
No symbols loaded for: version.dll

Command: Commands are comma separated (like assembly instructions): mov eax, ebx Default

Paused Breakpoint at 7617B30 set! Time Wasted Debugging: 0:00:13:10

Setting breakpoint at LoadLibraryW

CPU Log Notes Breakpoints Memory Map Call Stack SEH Script Symbols Source Referer

Base	Module	Address	Type	Ordinal	Symbol
75860000	win32u.dll				
73EA0000	version.dll	73EA509C	Import		kernelbase.LoadLibraryW
75A10000	user32.dll				
76EF0000	ucrtbase.dll				
76350000	sechost.dll				
757A0000	rpcrt4.dll				
00400000	random.exe				
76290000	oleaut32.dll				
755D0000	ole32.dll				
77730000	ntdll.dll				
77090000	msvcrt.dll				
75FD0000	msvc_p_win.dll				
76050000	kernelbase.dll				
77630000	kernel32.dll				
75E00000	gdi32full.dll				
75D10000	gdi32.dll				
6FF20000	comctl32.dll				
77380000	combase.dll				
75530000	apphelp.dll				
75EF0000	advapi32.dll				

Search: Type here to filter results... ☐ RegEx Search: LoadLibraryW ☐ Lock ☐ RegEx

No symbols loaded for: version.dll
[DIA] Skipping non-existent PDB: C:\Windows\SysWOW64\version.pdb
No symbols loaded for: version.dll

Command: Commands are comma separated (like assembly instructions): mov eax, ebx Default

Paused Breakpoint at 7618FB80 set! Time Wasted Debugging: 0:00:13:10

Setting breakpoint at CreateProcessW

The screenshot shows the WinDbg interface with the following components:

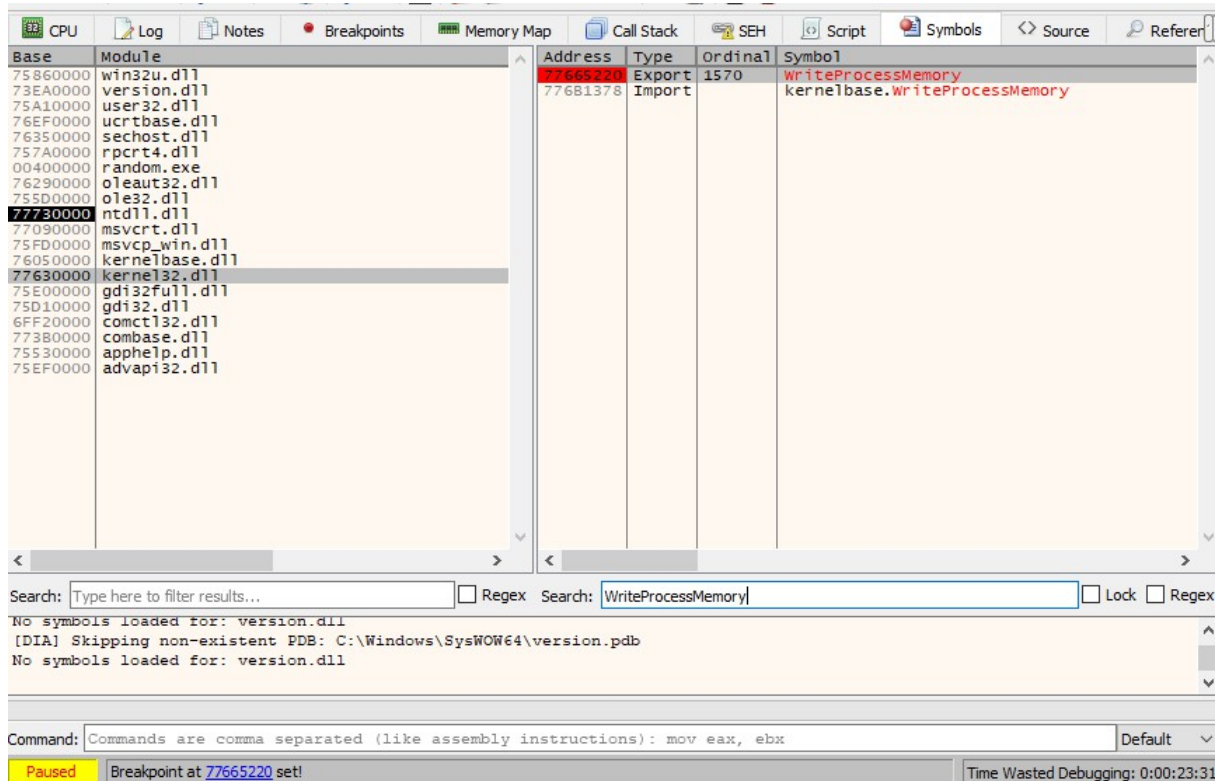
- Modules List:** A list of loaded modules with their base addresses. **kernel32.dll** is highlighted at address **77730000**.
- Symbols List:** A table showing symbols for the selected module. The symbol **kernel32.CreateProcessW** is selected at address **75AB93D4**.
- Search:** The search bar contains **CreatePr**. Below it, a message states: "No symbols loaded for: version.dll [DIA] Skipping non-existent PDB: C:\Windows\SysWOW64\version.pdb No symbols loaded for: version.dll".
- Command:** The command window shows **Commands are comma separated (like assembly instructions): mov eax, ebx**.
- Status Bar:** Displays **Paused**, **Breakpoint at 776488F0 set!**, and **Time Wasted Debugging: 0:00:17:44**.

Setting breakpoint at VirtualAlloc

The screenshot shows the WinDbg interface with the following components:

- Modules List:** A list of loaded modules with their base addresses. **kernelbase.dll** is highlighted at address **77730000**.
- Symbols List:** A table showing symbols for the selected module. The symbol **kernelbase.VirtualAlloc** is selected at address **77002164**.
- Search:** The search bar contains **VirtualAlloc**. Below it, a message states: "No symbols loaded for: version.dll [DIA] Skipping non-existent PDB: C:\Windows\SysWOW64\version.pdb No symbols loaded for: version.dll".
- Command:** The command window shows **Commands are comma separated (like assembly instructions): mov eax, ebx**.
- Status Bar:** Displays **Paused**, **Breakpoint at 76189100 set!**, and **Time Wasted Debugging: 0:00:21:02**.

Setting breakpoint at WriteProcessMemory



Verifying Breakpoints:

Type	Address	Module/Label/Exception	State	Disassembly	Hits
Software	776488E0	<kernel32.dll.CreateProcessW>	Enabled	mov edi,edi	0
	77665220	<kernel32.dll.WriteProcessMemory>	Enabled	mov edi,edi	0
	76171830	<kernelbase.dll.GetProcAddress>	Enabled	mov edi,edi	0
	76189100	<kernelbase.dll.VirtualAlloc>	Enabled	mov edi,edi	0
	7618F880	<kernelbase.dll.LoadLibraryW>	Enabled	mov edi,edi	0
	0047577C	<random.exe.OptionalHeader.Address01>	One-time	push ebp	0

Q5) If the sample unpacks or decrypts code/data at runtime, identify and extract at least one concrete artifact produced at runtime (e.g., decrypted string, embedded URL/IP, configuration blob, dropped file, or clearcode region in memory).

Ans) Hit Breakpoint upon running

Debugger interface showing CPU registers, memory dump, and assembly code. The assembly code is for the `kernelbase.dll!$121B30 #120F30 <GetProcAddress>` function. The registers show `EAX=76050000`, `EBX=00000000`, `ECX=76EF95A8`, `EDX=0019F7FC`, `EBP=0019F850`, `ESP=0019F83C`, `ESI=00000000`, and `EDI=77000F90`. The memory dump shows the function's code, including the `return to ucrtbase.wcsstr+B` instruction. The command line shows `INT3 breakpoint at <kernelbase.GetProcAddress> (76171B30)!`.

Debugger interface showing CPU registers, memory dump, and assembly code. The assembly code is for the `kernelbase.dll!$121B30 #120F30 <GetProcAddress>` function. The registers show `EAX=76050000`, `EBX=00000000`, `ECX=76EF95A8`, `EDX=0019F7FC`, `EBP=0019F850`, `ESP=0019F83C`, `ESI=00000000`, and `EDI=77000F90`. The memory dump shows the function's code, including the `return to ucrtbase.wcsstr+B` instruction. The command line shows `INT3 breakpoint at <kernelbase.GetProcAddress> (76171B30)!`.

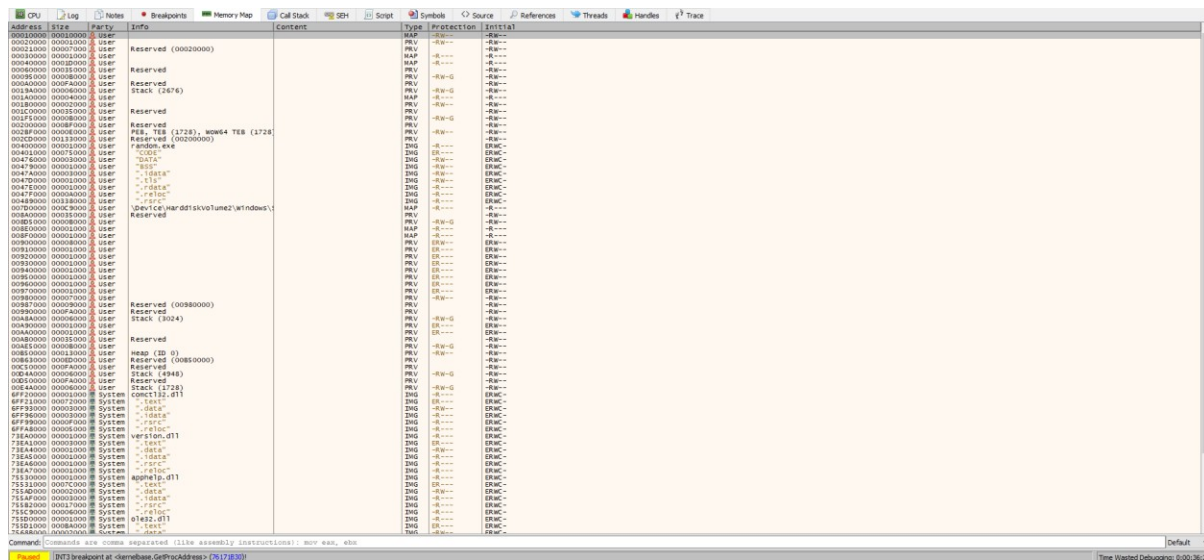
Call Stack Pane

Thread ID	Address	To	From	Size	Party	Comment
2676 - Main Thread	0019F83C	76F51977	kernelbase.GetProcAddress	18	System	
	0019F854	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F870	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F884	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F898	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F8B2	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F8C6	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F8D0	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F8E4	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F8F8	76F51A0D	ucrtbase.wcsstr+B	18	System	
1728	0019F90C	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F920	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F934	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F948	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F962	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F976	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F98A	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019F99E	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA10	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA24	76F51A0D	ucrtbase.wcsstr+B	18	System	
3024	0019FA38	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA52	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA66	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA80	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FA94	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAA8	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAB2	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAC6	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAD0	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAE4	76F51A0D	ucrtbase.wcsstr+B	18	System	
4948	0019FAD0	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAE4	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FAF8	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB12	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB26	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB40	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB54	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB68	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB82	76F51A0D	ucrtbase.wcsstr+B	18	System	
	0019FB96	76F51A0D	ucrtbase.wcsstr+B	18	System	

After Step over (F8)



Memory map:



Set Memory Breakpoint on Access to Write

random.exe - PID: 3920 - Module: kernelbase.dll - Thread: Main Thread 2676 - x32dbg [Elevated]

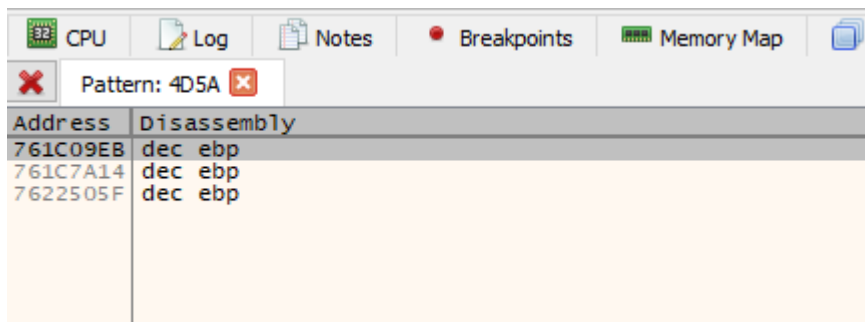
File View Debug Tracing Plugins Favourites Options Help Aug 19 2025 (TitanEngine)									
CPU Log Notes Breakpoints Memory Map Call Stack SEH Script Symbols Source									
Address	Size	Party	Info	Content	Type	Protection	Initial		
00010000	00010000	User	Reserved (00020000)		MAP	-RW--	-RW--		
00020000	00001000	User			PRV	-RW--	-RW--		
00021000	00007000	User			PRV	-RW--	-RW--		
00030000	00001000	User	Reserved		MAP	-R---	-R---		
00040000	0001D000	User			MAP	-R---	-R---		
00060000	00035000	User			PRV	-RW-G	-RW--		
00095000	00003000	User	Reserved		PRV	-RW--	-RW--		
00098000	00008000	User			PRV	-RW--	-RW--		
000A0000	000FA000	User			PRV	-RW-G	-RW--		
0019A000	00002000	User	Stack (2676)		PRV	-RW--	-RW--		
0019C000	00004000	User			MAP	-R---	-R---		
001A0000	00004000	User			PRV	-RW--	-RW--		
001B0000	00002000	User	Reserved		PRV	-RW--	-RW--		
001C0000	00035000	User			PRV	-RW-G	-RW--		
001F5000	00003000	User			PRV	-RW--	-RW--		
001F8000	00008000	User	Reserved		PRV	-RW--	-RW--		
00200000	0000F000	User			PRV	-RW--	-RW--		
002B0000	0000E000	User			PRV	-RW--	-RW--		
002CD000	00133000	User	random.exe		PRV	-R---	-R---		
00400000	00001000	User		"CODE"	IMG	ER---	ERWC-		
00401000	00075000	User		"DATA"	IMG	-RW--	ERWC-		
00476000	00001000	User	"BSS", ".idata", ".tls"		IMG	-RW--	ERWC-		
00477000	00002000	User			IMG	-RW--	ERWC-		
00479000	00005000	User			IMG	-RW--	ERWC-		
0047E000	00343000	User	"rdata", ".reloc", ".rsrc"		IMG	-R---	ERWC-		
007D0000	000C9000	User			MAP	-R---	-R---		
008A0000	00035000	User			PRV	-RW-G	-RW--		
008D5000	00003000	User	Reserved		PRV	-RW--	-RW--		
008D8000	00008000	User			MAP	-R---	-R---		
008E0000	00001000	User			MAP	-R---	-R---		
008F0000	00001000	User	Reserved (00980000)		PRV	-RW--	-RW--		
00900000	00008000	User			PRV	-RW--	-RW--		
00910000	00001000	User			PRV	-RW--	-RW--		
00920000	00001000	User	Reserved		PRV	-RW--	-RW--		
00930000	00001000	User			PRV	-RW--	-RW--		
00940000	00001000	User			PRV	-RW--	-RW--		
00950000	00001000	User	Stack (3024)		PRV	-RW-G	-RW--		
00960000	00001000	User			PRV	-RW--	-RW--		
00970000	00001000	User			PRV	-RW--	-RW--		
00980000	00007000	User	Reserved		PRV	-RW--	-RW--		
00987000	00009000	User			PRV	-RW--	-RW--		
00990000	000FA000	User			PRV	-RW-G	-RW--		
00A8A000	00002000	User	Stack (4948)		PRV	-RW--	-RW--		
00A8C000	00004000	User			PRV	-RW--	-RW--		
00A90000	00001000	User			PRV	-RW--	-RW--		
00AA0000	00001000	User	Reserved		PRV	-RW--	-RW--		
00AB0000	00035000	User			PRV	-RW--	-RW--		
00AE5000	00003000	User			PRV	-RW--	-RW--		
00AE8000	00008000	User	Heap (ID 0)		PRV	-RW--	-RW--		
00B50000	00013000	User			PRV	-RW--	-RW--		
00B63000	000ED000	User			PRV	-RW--	-RW--		
00C50000	000FA000	User	Reserved		PRV	-RW--	-RW--		
00D4A000	00002000	User			PRV	-RW-G	-RW--		
00D4C000	00004000	User			PRV	-RW--	-RW--		
00D50000	000FA000	User	Stack (1728)		PRV	-RW--	-RW--		
00E4A000	00002000	User			PRV	-RW--	-RW--		
00E4C000	00004000	User			PRV	-RW--	-RW--		
6FF20000	00001000	System	comctl32.dll		IMG	-R---	ERWC-		
6FF21000	00072000	System	"text"		IMG	ER---	ERWC-		
6FF93000	00001000	System	"data"		IMG	-RW--	ERWC-		
6FF94000	00002000	System	"data"		IMG	-RW--	ERWC-		
6FF96000	00017000	System	"idata", ".rsrc", ".reloc"		IMG	-R---	ERWC-		
73EA0000	00001000	System	version.dll		IMG	-R---	ERWC-		
73EA1000	00003000	System	"text"		IMG	ER---	ERWC-		
73EA4000	00001000	System	"data"		IMG	-RW--	ERWC-		
73EA5000	00003000	System	"idata", ".rsrc", ".reloc"		IMG	-R---	ERWC-		
75530000	00001000	System	apphelp.dll		IMG	-R---	ERWC-		
75531000	0007C000	System	"text"		IMG	ER---	ERWC-		
755AD000	00002000	System	"data"		IMG	-RW--	ERWC-		
755AF000	00020000	System	"idata", ".rsrc", ".reloc"		IMG	-R---	ERWC-		
755D0000	00001000	System	ole32.dll		IMG	-R---	ERWC-		
755D1000	0008A000	System	"text"		IMG	ER---	ERWC-		
75688000	00001000	System	"data"		IMG	-RW--	ERWC-		

Command: Commands are comma separated (like assembly instructions): mov eax, ebx

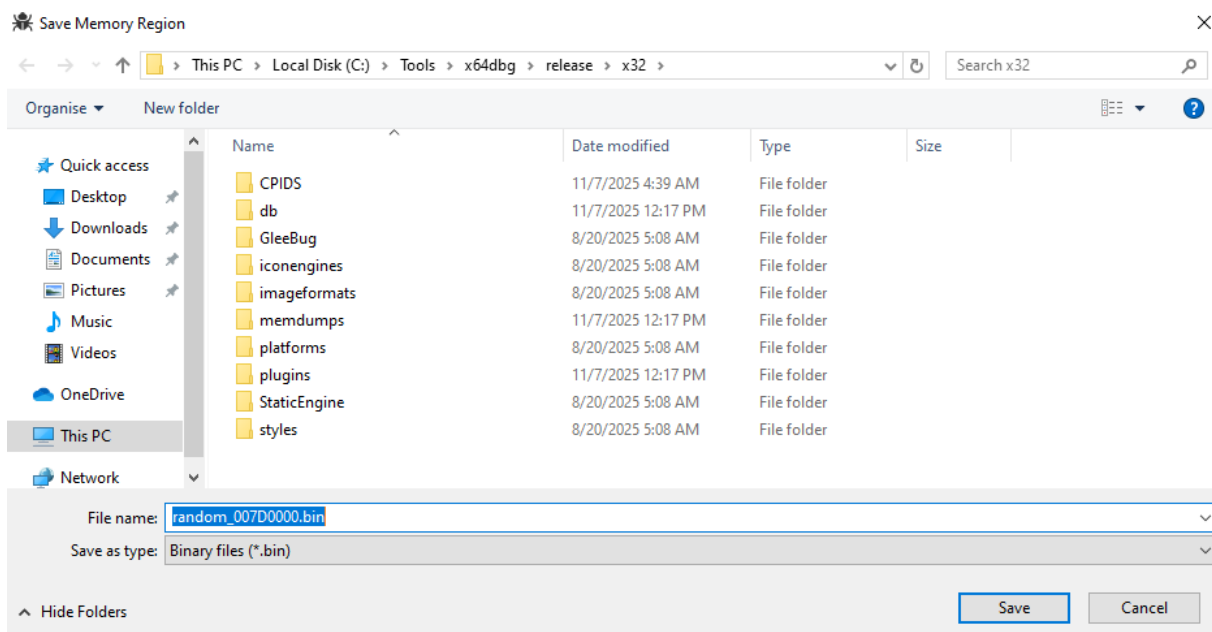
Paused Memory breakpoint at 007D0000[000C9000] set!

OEP:

File View Debug Tracing Plugins Favourites Options Help Aug 19 2025 (TitanEngine)									
CPU Log Notes Breakpoints Memory Map Call Stack SEH Script Symbols Source									
76171000	8B4C	mov	ebp, esp						
76171001	8B4C	mov	ebp, esp						
76171002	8B4C	mov	ebp, esp						
76171003	8B4C	mov	ebp, esp						
76171004	8B4C	mov	ebp, esp						
76171005	8B4C	mov	ebp, esp						
76171006	8B4C	mov	ebp, esp						
76171007	8B4C	mov	ebp, esp						
76171008	8B4C	mov	ebp, esp						
76171009	8B4C	mov	ebp, esp						
7617100A	8B4C	mov	ebp, esp						
7617100B	8B4C	mov	ebp, esp						
7617100C	8B4C	mov	ebp, esp						
7617100D	8B4C	mov	ebp, esp						
7617100E	8B4C	mov	ebp, esp						
7617100F	8B4C	mov	ebp, esp						
76171010	8B4C	mov	ebp, esp						
76171011	8B4C	mov	ebp, esp						
76171012	8B4C	mov	ebp, esp						
76171013	8B4C	mov	ebp, esp						
76171014	8B4C	mov	ebp, esp						
76171015	8B4C	mov	ebp, esp						
76171016	8B4C	mov	ebp, esp						
76171017	8B4C	mov	ebp, esp						
76171018	8B4C	mov	ebp, esp						
76171019	8B4C	mov	ebp, esp						
7617101A	8B4C	mov	ebp, esp						
7617101B	8B4C	mov	ebp, esp						
7617101C	8B4C	mov	ebp, esp						
7617101D	8B4C	mov	ebp, esp						
7617101E	8B4C	mov	ebp, esp						
7617101F	8B4C	mov	ebp, esp						
76171020	8B4C	mov	ebp, esp						
76171021	8B4C	mov	ebp, esp						
76171022	8B4C	mov	ebp, esp						
76171023	8B4C	mov	ebp, esp						
76171024	8B4C	mov	ebp, esp						
76171025	8B4C	mov	ebp, esp						
76171026	8B4C	mov	ebp, esp						
76171027	8B4C	mov	ebp, esp						
76171028	8B4C	mov	ebp, esp						
76171029	8B4C	mov	ebp, esp						
7617102A	8B4C	mov	ebp, esp						
7617102B	8B4C	mov	ebp, esp						
7617102C	8B4C	mov	ebp, esp						
7617102D	8B4C	mov	ebp, esp						
7617102E	8B4C	mov	ebp, esp						
7617102F	8B4C	mov	ebp, esp						
76171030	8B4C	mov	ebp, esp						
76171031	8B4C	mov	ebp, esp						
76171032	8B4C	mov	ebp, esp						
76171033	8B4C	mov	ebp, esp						
76171034	8B4C	mov	ebp, esp						
76171035	8B4C	mov	ebp, esp						
76171036	8B4C	mov	ebp, esp						
76171037	8B4C	mov	ebp, esp						
76171038	8B4C	mov	ebp, esp						
76171039	8B4C	mov	ebp, esp						
7617103A	8B4C	mov	ebp, esp						
7617103B	8B4C	mov	ebp, esp						
7617103C	8B4C	mov	ebp, esp						
7617103D	8B4C	mov	ebp, esp						
7617103E	8B4C	mov	ebp, esp						
7617103F	8B4C	mov	ebp, esp						
76171040	8B4C	mov	ebp, esp						
76171041	8B4C	mov	ebp, esp						
76171042	8B4C	mov	ebp, esp						
76171043	8B4C	mov	ebp, esp						
76171044	8B4C	mov	ebp, esp						
76171045	8B4C	mov	ebp, esp						
76171046	8B4C	mov	ebp, esp						
76171047	8B4C	mov	ebp, esp						
76171048	8B4C	mov	ebp, esp						
76171049	8B4C	mov	ebp, esp						
7617104A	8B4C	mov	ebp, esp						
7617104B	8B4C	mov	ebp, esp						
7617104C									



Dumping the file



Q6) Write a short conclusion (3–4 sentences) summarizing the sample's likely intent (e.g., downloader, dropper, info-stealer, persistence) and give one concrete mitigation/recovery

recommendation that an IT team should apply if a similar binary is found on the network

(e.g., block C2 domains/IPs, isolate affected hosts, remove scheduled tasks, rotate credentials).

Ans) Conclusions that we can draw from this analysis:

Based on the behavioral indicators which include dynamic API resolution (LoadLibrary / GetProcAddress), remote memory writing (WriteProcessMemory), thread injection (CreateRemoteThread) and suspicious memory allocations

(VirtualAlloc / VirtualProtect) the sample is highly likely to be a dropper/injector designed to unpack or load additional malicious payloads into memory.

These characteristics suggest an intent to evade static analysis and execute a secondary stage (potentially ransomware or credential-stealing malware). The use of process injection and dynamic imports indicates deliberate efforts to bypass detection.

Mitigation / Recovery Recommendation:

Immediately isolate the affected machine from the network and perform credential rotation for any accounts used from that system. In parallel, block any observed C2 IP/domains at the firewall and remove any persistence mechanisms.